

PLA - Programmable Logic Array

→ It is fixed architecture logic device with programmable AND gates and programmable OR gates.

- Steps to program PLA



1:17 / 9:17

Designing Steps of PLA >



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- Steps to program PLA

Step-1 Find boolean function from truth table.

A	B	C	f_1	f_2
0	0	0	0	0
0	0	1	0	0
0	1	0	0	1
0	1	1	1	0
1	0	0	0	0
1	0	1	1	1
1	1	0	0	0
1	1	1	1	1

f_1

A	BC			
	00	01	11	10
0	0	0	1	0
1	0	1	1	0

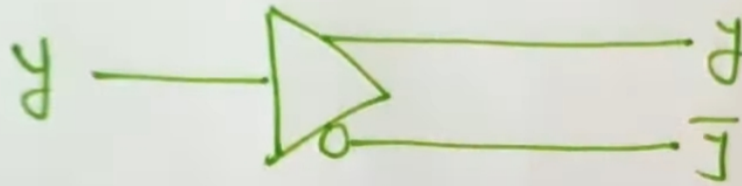
$$f_1 = AC + BC$$

$$J_2 = AC + \bar{A}B\bar{C}$$

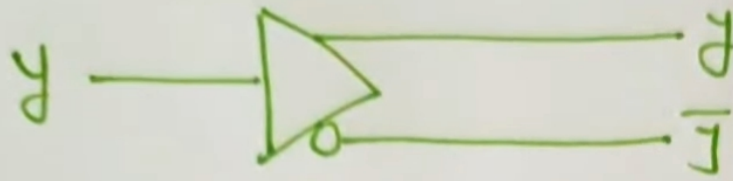
Step-2 Identity no of Input buttons.

- we have three Input variables. So, we need

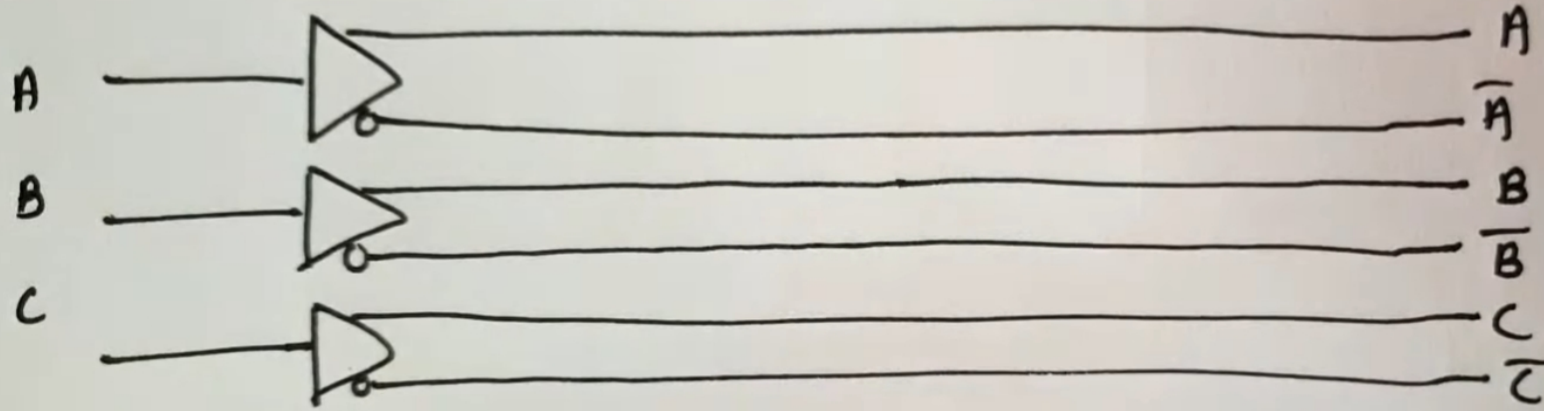
No of Input buttons = 3



Step-3 Implimentation of b



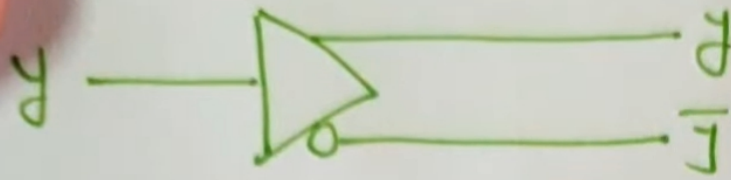
Step-3 Implementation of boolean function in PLA



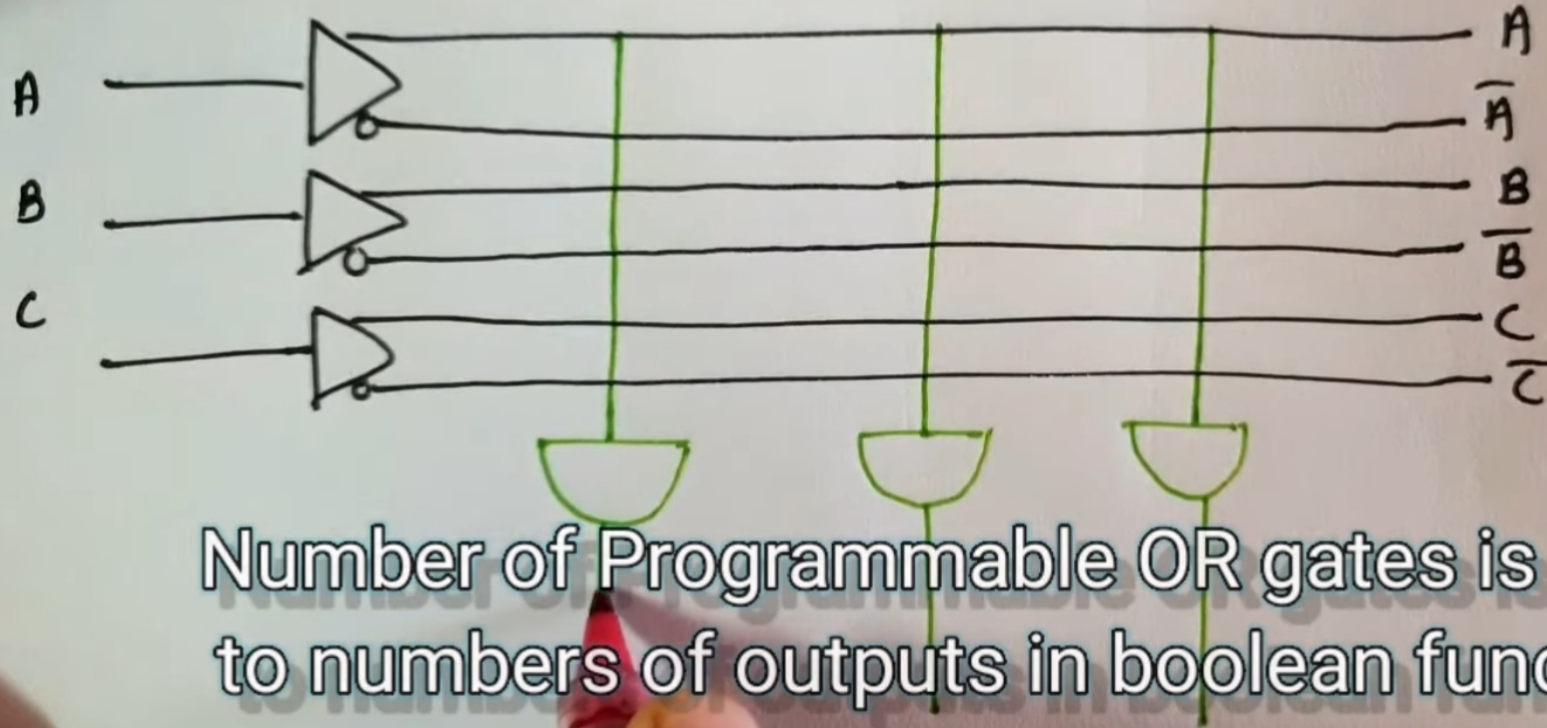
$$Y_1 = AC + BC$$

$$Y_2 = AC + \bar{A}\bar{B}$$

Number of Programmable AND gates is equals to numbers of product terms in boolean function.



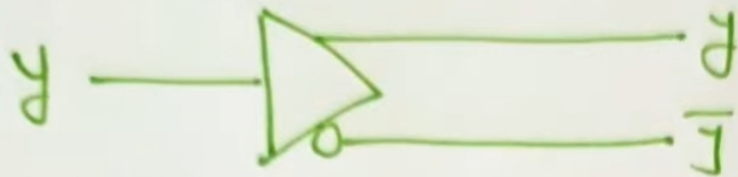
Step-3 Implimentation of boolean function in PLA



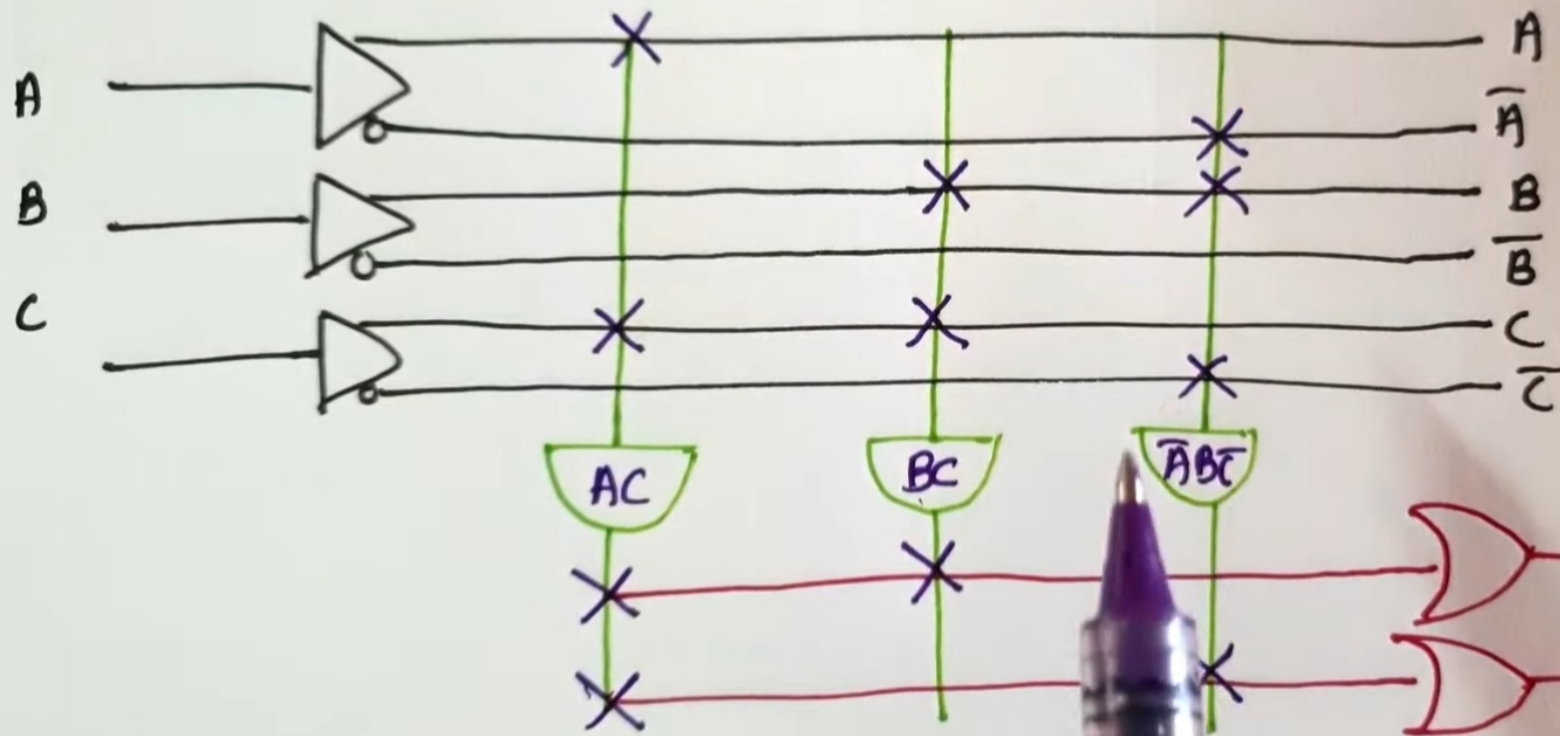
$$y_1 = AC + BC$$

$$y_2 = AC + \bar{A}B\bar{C}$$

Number of Programmable OR gates is equals to numbers of outputs in boolean functions.



Step-3 Implementation of boolean function in PLA



$$f_1 = AC + BC$$

$$f_2 = AC + \bar{A}\bar{B}\bar{C}$$

$$AC + BC$$

$$f_1 =$$

$$f_2 =$$

$$AC + \bar{A}\bar{B}\bar{C}$$

PAL - Programmable Array Logic

Programmable Array Logic architecture includes

1. Programmable AND gates
2. Fixed OR gates

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→ It is most commonly used PLD.

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It provides simple structure

It has issues regarding flexibility.

$$X = \sum m(2, 3, 6, 7)$$

$$Y = \sum m(0, 2, 3, 5)$$

$$Z = \sum m(1, 6, 7)$$

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X \ A \ BC				
	00	01	11	10
0			1	1
1			1	1

$$X = B$$

Z \ A \ BC				
	00	01	11	10
0		1		
1			1	1

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It has issues regarding flexibility.

Y \ A \ BC				
	00	01	11	10
0	1		1	1
1		1		

$$Y = \bar{A}B + \bar{A}\bar{C} + A\bar{B}C$$

$$Z = AB + \bar{A}\bar{B}C$$

$$X = B$$

$$Y = \bar{A}B + \bar{A}\bar{C} + A\bar{B}C$$

Z	BC		11	10
	00	01		
A				
0		1		
1			1	1

$$Z = AB + \bar{A}\bar{B}C$$

